Ayush Ramashankar Yadav

Harrison, NJ | ayushyadav1783@gmail.com | github.com/ayushyadav1783-cmd | linkedin.com/in/ayush-yadav-87ab581a5

Computer Vision & Robotics Researcher *— built a 29.6 dB language-grounded 3D Gaussian Splatting pipeline for real-robot manipulation at NJIT's Aegean AI Lab.*

Education

New Jersey Institute of Technology (NJIT) **Newark, NJ**
M.S. in Artificial Intelligence | GPA: 3.83/4.00 | Computer Vision, Deep Learning, AI for Robotics, ML *2025 – 2027 (Expected)*

Ramrao Adik Institute of Technology (D.Y. Patil University) **India**
B.Tech. in Computer Engineering | CGPA: 8.66/10.00 *2021 – 2025*

Research Experience

Aegean AI Lab, NJIT **Newark, NJ**
AI Research Assistant — Summer of Agents 2026 (competitive program, 3 positions) *Jun 2026 – Present*

Language-Grounded 3D Gaussian Splatting for Robotic Manipulation (AR4 arm, Real2Sim2Real):

- Building an end-to-end pipeline from natural-language query to robot action: text → CLIP-embedded 3DGS scene → 3D object localization → XYZ manipulation targets for a SmolVLA motion planner on a physical AR4 arm
- Achieved 29.60 dB PSNR / 0.952 SSIM / 0.078 LPIPS lab-scene reconstruction (621/621 gimbal-captured frames registered at 0.96px), a +2.1 dB improvement over baseline, trained with gsplat MCMC on an NVIDIA RTX PRO 6000 Blackwell (96GB)
- Produced two documented ablation findings: per-folder COLMAP camera mode for mixed video/still captures (14 dB → 29 dB on identical images) and Gaussian budget vs. training length (1M @ 30K steps outperforms 3M @ 100K)
- Diagnosed a catastrophically broken scene scale (∼5M units) traced to sequential matching on unordered captures; root-caused it and resolved it via exhaustive feature matching

Sim-to-Real Geometric Alignment (code-reviewed and merged to aegean-ai/ar4-physical-ai):

- Built and validated a Umeyama similarity-transform engine (sub-mm synthetic accuracy) and a scale-aware robust ICP engine with FPFH+RANSAC global initialization (0.46mm on synthetic ground truth) for 3DGS-to-robot-frame registration
- Found that unconstrained 7-DOF ICP silently converges to a 28° mis-tilt (5.8mm RMSE) on quasi-symmetric geometry; designed a gravity-constrained pipeline (table-plane leveling, closed-form yaw/scale/translation, 12-seed yaw sweep) proving low RMSE alone cannot validate registration

Open-Vocabulary Semantic 3D Queries:

- Engineered CLIP ViT-B/32 feature distillation into trained Gaussians with a semantic query engine, achieving 9/10 correct 3D object localizations on real captures (4/5 open-vocabulary, beyond closed-set detectors)
- Built a ROS2+Gazebo autonomous capture pipeline (304 keyframes) feeding sim-side training data collection

Projects

Faster R-CNN Hyperparameter Optimization | PyTorch, Optuna, W&B, COCO, VisDrone **Feb 2026**

- Built a 4-stage Optuna HPO pipeline (training, RPN anchors, RoI sampling, inference) on COCO-2017; cut per-trial runtime 30 min → 8 min via deferred evaluation and improved tuned mAP 5% over baseline on VisDrone with a custom VOC-to-COCO converter

Robotics Semantic Perception Pipeline | ROS2, YOLOv8, Zenoh, PostgreSQL, pgvector, Apache AGE **Mar 2026**

- Built an end-to-end sensor-to-database pipeline (TurtleBot3/Gazebo → ROS2 → YOLOv8 → Zenoh → PostgreSQL) ingesting 500+ detection events; extended with 512-dim embeddings in pgvector and a semantic graph enabling object re-localization

UAV Drone Detection & Multi-Object Tracking | YOLOv8, Kalman Filter, OpenCV **Mar 2026**

- Fine-tuned YOLOv8n on 3,000+ drone images; built a custom 8-state Kalman tracker [cx, cy, w, h, vx, vy, vw, vh] with Hungarian-algorithm assignment and a full annotation pipeline overlaying tracks, predictions, and trajectory polylines

Temporal Cycle-Consistency Self-Supervised Video Learning | PyTorch **Mar 2026**

- Implemented TCC loss and temporal segmentation from scratch; swept embedding dimensions D={32, 64, 128} with D=128 achieving best alignment (0.589), analyzed via PCA, UMAP, and t-SNE visualizations

Technical Skills

Languages: Python, C/C++, SQL, Java, Bash

ML/DL Frameworks: PyTorch, Transformers, Hugging Face, OpenCV, torchvision, CUDA (mixed-precision & multi-GPU training)

Computer Vision & 3D Vision: CLIP, YOLOv8, Faster R-CNN, COCO API, object detection & tracking, segmentation, GradCAM/attention visualization, VideoMAE, self-supervised learning (TCC), 3D Gaussian Splatting (gsplat), COLMAP, point-cloud registration (ICP, Umeyama, FPFH+RANSAC)

Robotics & Systems: ROS2, Gazebo, Zenoh, sensor fusion

MLOps & Data: Weights & Biases, Optuna, Docker, Git/GitHub Actions, AWS (EC2, S3), PostgreSQL, pgvector, vector databases